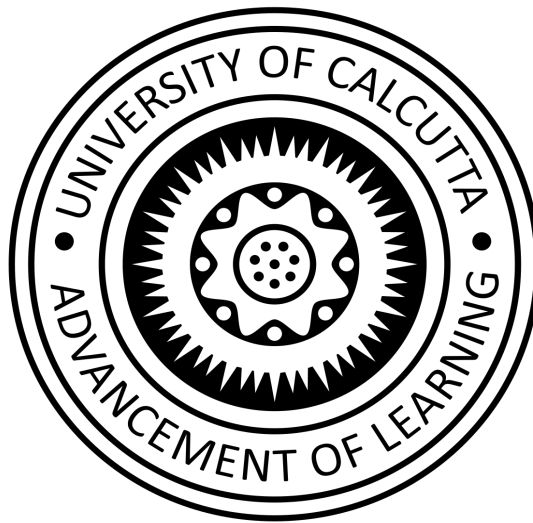


# **MUSIC LISTENING HABITS AND PSYCHOLOGICAL WELLBEING IN YOUNG ADULTS**

**A Case Study**



**University of Calcutta**

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## **ABSTRACT**

In this project a survey was conducted among 315 individuals. The questionnaire assessed uses of music in everyday life: frequency of listening, situations where music is encountered, emotional responses to music, and motives for listening (i.e., listening strategies). Also, different facets of psychological well-being were assessed. Results showed that listening to music is a common leisure activity encountered in many everyday situations, and that listening to music is a frequent source of positive emotions for older adults. Also, the participants reported using a variety of listening strategies related to emotional functions (e.g., pleasure, mood regulation, and relaxation) and issues of identity, belonging, and agency. The associations between listening strategies and well-being were explored through correlation and multiple regression analyses where the influence of background variables was controlled for. Health status and personality were the most important predictors of well-being, but some listening strategies were also significantly associated with psychological well-being. The results give important insights into older adults' uses of music in everyday life and give clues regarding possible relationships between musical activities and well-being.

**KEYWORDS:** Emotion, everyday life, identity, music, psychological well-being, Gender, Diagrams, Regression, Binary, Logistic, Outlier, Significant, Principal Component, Ordinal, Simpson's Paradox, Hypothesis.

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## **1. INTRODUCTION**

**“Music gives a soul to the universe, wings to the mind, flight to the imagination, and life to everything”**

What are the connections between us and music? The answers include: the pleasure gained from listening; the warmth and friendship from being part of a group making music; the stimulus and satisfaction from regular practice and rehearsal; the intellectual delight from exploring the intricacies of musical forms and structures; the physical energy released within us by both playing and listening to music, inspiring us often to move and dance.

There are at least three ways how people can relate to music namely, as a composer, as a performer (by means of their voice or an instrument), and as a listener.

Music psychology is related to music listening in several respects which will accompany this whole work: Why do we listen to music at all? What kind of music do we listen to and why? How strong is our relation to specific music and why? What are our healing experiences because of listening to some music? What is the role of the context in which the music is heard?

The value one places on music in everyday life depends on the uses they make of it and the degree of engagement with it, which is dependent on the context in which the music is heard. ‘Which music do we like?’ has become one of the most often used questions in psychological research, for two reasons: First, since every person is in contact with music every day and most people see music as one of the most important things in their daily life, the study of music listening provides an authentic and fruitful avenue to their experiences and behavior in a variety of situations and contexts. Second, because music has been recognized as a tool for expressing and inducing moods and emotions and also as a means to convey information in social environments, music is by now often used as an essential implement in personality, emotion, and social psychology research.

Thus, research on music preference is going on to provide us with deep insights into many psychological questions.

However, there is very little research that progresses beyond the purely speculative when it comes to explaining why music should have an effect on wellbeing. This study proposes that music listening may have an effect on well-being because it satisfies basic psychological aims, and presents an investigation of everyday music listening and its associations with psychological well-being in a sample of people.

Strong evidence from a variety of sources indicates that music is capable of inducing effect in listeners (for a review, see Juslin and Laukka, 2004). For instance, questionnaire and diary studies confirm that listeners actively use music for mood regulation and that music induces positive affective states (e.g., North et al., 2004; Thayer et al., 1994). Brain imaging studies suggest that listeners' brain responses to music involve the same regions that are involved in reward and pleasure (e.g., Blood and Zatorre, 2001; Menon and Levitin, 2005). Studies of peak experiences indicate that music is one of the most effective triggers of such psychological states (Panzarella, 1980), and that peak experiences with music often involve strongly felt positive emotions (Gabrielsson, 2001). Mood induction studies further indicate that music may be one of the most effective mood induction strategies available (Va štija ĩl, 2002), and music is also considered an effective means for decreasing stress-related arousal (Pelletier, 2004). Though the emotional functions of music have received most attention, music may also serve other psychological aims (see DeNora, 2000; Lull, 1987; Ruud, 1997). For instance, music is 216 PETRI LAUKKA regarded to play an important role in the construction and strengthening of identity and self. Further, music can help to enhance social communication and peer group identification, as well as provide comfort and reduce loneliness.

Finally, personal ‘agency’ refers to the feeling that one can be the cause of events (i.e., of being the one who acts in one’s own l3.1 )

## **2. DATASET**

A simple still reasonable connection is comprehended between Music listening habits and psychological wellbeing among

students of age 15-25 in our dataset. Since Gender plays a dominant role in understanding the psychological mind of a person; the dataset has been tactfully divided between two genders – Female and Male.

The first few data points of our data set are tabulated below:

**Table-1: First six data points**

| <b>What is your age?</b>                  | <b>What is your MOTHER TONGUE?</b>    | <b>Gender</b>                  | <b>What is your Zodiac/Star Sign?</b> | <b>Are you a MUSICIAN?</b>         |
|-------------------------------------------|---------------------------------------|--------------------------------|---------------------------------------|------------------------------------|
| 20                                        | Bengali                               | Female                         | Virgo                                 | No                                 |
| 18                                        | Bengali                               | Male                           | Capricorn                             | Yes                                |
| 20                                        | Bengali                               | Female                         | Virgo                                 | Yes                                |
| 20                                        | Bengali                               | Female                         | Virgo                                 | No                                 |
| 21                                        | Bengali                               | Female                         | Taurus                                | No                                 |
| 21                                        | Bengali                               | Female                         | Pisces                                | Yes                                |
| <b>In which language do you prefer to</b> | <b>How many hours in a day do you</b> | <b>Mostly in which time do</b> | <b>Mostly in which time do you</b>    | <b>Mostly in which time do you</b> |

| <b>listen to music?</b>                   | <b>listen to music?</b>            | <b>you prefer to listen to music? [Rock]</b> | <b>prefer to listen to music? [Classical]</b> | <b>prefer to listen to music[Slow +Reverb (Lofi)]</b> |
|-------------------------------------------|------------------------------------|----------------------------------------------|-----------------------------------------------|-------------------------------------------------------|
| Bengali, English, Hindi                   | 3.6 Hrs.                           |                                              | Evening                                       | Morning                                               |
| Bengali, English, Hindi                   | 1.6 Hrs.                           |                                              | Morning                                       |                                                       |
| Bengali, English, Hindi, Tamil, Telegu    | 1 hour - 3 hour                    |                                              | Morning                                       | Night and late night                                  |
| Hindi                                     | 2.6 Hrs.                           |                                              | Evening                                       | Night and late night                                  |
| English                                   | 0.3 Hrs.                           | Morning                                      | Afternoon                                     | Evening                                               |
| Bengali, Hindi                            | 0.8 Hrs.                           |                                              | Afternoon                                     |                                                       |
| <b>Mostly in which time do you prefer</b> | <b>Mostly in which time do you</b> | <b>Mostly in which time do</b>               | <b>Mostly in which time do you</b>            | <b>Mostly in which time do you</b>                    |

| <b>to listen to music?</b><br><b>Instrumental</b>                                             | <b>prefer to listen to music?</b><br><b>HipHop</b>                                           | <b>you prefer to listen to music?</b><br><b>Folk</b>                                                 | <b>prefer to listen to music?</b><br><b>Romantic</b>                                                  | <b>prefer to listen to music?</b><br><b>Devotional</b>                                                |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Night and Late night                                                                          | Afternoon                                                                                    |                                                                                                      |                                                                                                       |                                                                                                       |
| Night and Late night                                                                          |                                                                                              | Evening                                                                                              | Afternoon                                                                                             |                                                                                                       |
|                                                                                               |                                                                                              |                                                                                                      | Evening                                                                                               |                                                                                                       |
|                                                                                               |                                                                                              | Afternoon                                                                                            | Morning                                                                                               |                                                                                                       |
| Night and Late night                                                                          |                                                                                              |                                                                                                      |                                                                                                       |                                                                                                       |
|                                                                                               |                                                                                              | Evening                                                                                              | Night and Late night                                                                                  | Morning                                                                                               |
| <b>How often do you listen to music because of the following reasons? [For entertainment]</b> | <b>How often do you listen to music because of the following reasons? [It evokes memory]</b> | <b>How often do you listen to music because of the following reasons? [Interest in music itself]</b> | <b>How often do you listen to music because of the following reasons? [Enjoy to listen to lyrics]</b> | <b>How often do you listen to music because of the following reasons? [To enhance positive moods]</b> |

|                                                                                                                           |                                                                                                                                            |                                                                                                                                                |                                                                                                                                           |                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Sometimes<br>(3)                                                                                                          | Often (4)                                                                                                                                  | Always (5)                                                                                                                                     | Sometimes<br>(3)                                                                                                                          | Often (4)                                                                                                                               |
| Often (4)                                                                                                                 | Sometimes<br>(3)                                                                                                                           | Always (5)                                                                                                                                     | Always (5)                                                                                                                                | Often (4)                                                                                                                               |
| Sometimes<br>(3)                                                                                                          | Often (4)                                                                                                                                  | Always (5)                                                                                                                                     | Often (4)                                                                                                                                 | Sometimes<br>(3)                                                                                                                        |
| Sometimes<br>(3)                                                                                                          | Always (5)                                                                                                                                 | Often (4)                                                                                                                                      | Always (5)                                                                                                                                | Often (4)                                                                                                                               |
| Sometimes<br>(3)                                                                                                          | Sometimes<br>(3)                                                                                                                           | Sometim-<br>es (3)                                                                                                                             | Sometimes<br>(3)                                                                                                                          | Sometimes<br>(3)                                                                                                                        |
| Often (4)                                                                                                                 | Often (4)                                                                                                                                  | Often (4)                                                                                                                                      | Often (4)                                                                                                                                 | Often (4)                                                                                                                               |
| <b>How often do<br/>you listen to<br/>music<br/>because of<br/>the following<br/>reasons? [ To<br/>ease<br/>pressure]</b> | <b>How often<br/>do you<br/>listen to<br/>music<br/>because of<br/>the<br/>following<br/>reasons?<br/>[To lessen<br/>your<br/>boredom]</b> | <b>How<br/>common<br/>is it that<br/>you feel<br/>each of<br/>the<br/>following<br/>emotions<br/>in<br/>response<br/>to music?<br/>[Happy]</b> | <b>How<br/>common is<br/>it that you<br/>feel each of<br/>the<br/>following<br/>emotions in<br/>response to<br/>music?<br/>[Enjoying]</b> | <b>How<br/>common is<br/>it that you<br/>feel each of<br/>the<br/>following<br/>emotions in<br/>response to<br/>music?<br/>[Loving]</b> |

|                                                                                                          |                                                                                                      |                                                         |                                                    |                                                       |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------|
| Often (4)                                                                                                | Always (5)                                                                                           | 4. Always                                               | 4. Always                                          | 3. Often                                              |
| Sometimes (3)                                                                                            | Often (4)                                                                                            | 3. Often                                                | 4. Always                                          | 4. Always                                             |
| Sometimes (3)                                                                                            | Seldom (2)                                                                                           | 3. Often                                                | 3. Often                                           | 3. Often                                              |
| Often (4)                                                                                                | Always(5)                                                                                            | 3. Often                                                | 3. Often                                           | 4. Always                                             |
| Seldom (2)                                                                                               | Seldom (2)                                                                                           | 3. Often                                                | 4. Always                                          | 2. Seldom                                             |
| Often (4)                                                                                                | Often (4)                                                                                            | 3. Often                                                | 3. Often                                           | 3. Often                                              |
| <b>How common is it that you feel each of the following emotions in response to music?<br/>Enchanted</b> | <b>How common is it that you feel each of the following emotions in response to music?<br/>[Sad]</b> | <b>Some aspects of your personality<br/>Self Loving</b> | <b>Some aspects of your personality<br/>Daring</b> | <b>Some aspects of your personality<br/>Introvert</b> |
| 2. Seldom                                                                                                | 3. Often                                                                                             | Yes                                                     | Yes                                                | Yes                                                   |
| 3. Often                                                                                                 | 2. Seldom                                                                                            | No                                                      | Yes                                                | No                                                    |

|                                                               |                                                                    |                                                                                                |                                                                                             |                                                                                             |
|---------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 2. Seldom                                                     | 3. Often                                                           | Yes                                                                                            | No                                                                                          | Yes                                                                                         |
| 2. Seldom                                                     | 2. Seldom                                                          | Yes                                                                                            | Yes                                                                                         | No                                                                                          |
| 4. Always                                                     | 3. Often                                                           | Yes                                                                                            | Yes                                                                                         | Yes                                                                                         |
| 3. Often                                                      | 3. Often                                                           | Yes                                                                                            | No                                                                                          | Yes                                                                                         |
| <b>Some aspects<br/>of your<br/>personality<br/>Emotional</b> | <b>Some<br/>aspects of<br/>your<br/>personality<br/>Optimistic</b> | <b>Which<br/>app do<br/>you use<br/>the most<br/>for<br/>listening<br/>to music?<br/>[1st]</b> | <b>Which app<br/>do you use<br/>the most<br/>for<br/>listening to<br/>music ?<br/>[2nd]</b> | <b>Which app<br/>do you use<br/>the most<br/>for<br/>listening to<br/>music ?<br/>[3rd]</b> |
| No                                                            | Yes                                                                | Spotify                                                                                        | Gaana                                                                                       | Jiosaavn                                                                                    |
| Yes                                                           | Yes                                                                | Spotify                                                                                        | Youtube                                                                                     | Gaana                                                                                       |
| Yes                                                           | Yes                                                                | Gaana                                                                                          | Jiosaavn                                                                                    | Youtube                                                                                     |
| Yes                                                           | No                                                                 | Youtube                                                                                        | Gaana                                                                                       | Jiosaavn                                                                                    |
| Yes                                                           | Yes                                                                | Spotify                                                                                        | Youtube                                                                                     | Gaana                                                                                       |
| No                                                            | Yes                                                                | Youtube                                                                                        | Gaana                                                                                       | Spotify                                                                                     |

|  | When you don't have an internet connection then in what way do you get indulged in music? | Do you listen to music while working? | Does music has any effect on you |  |
|--|-------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------|--|
|  | TV                                                                                        | yes                                   | Yes                              |  |
|  | Singing songs                                                                             | yes                                   | Yes                              |  |
|  | Singing songs                                                                             | yes                                   | Yes                              |  |
|  | Downloading music                                                                         | yes                                   | Yes                              |  |
|  | Downloading music                                                                         | yes                                   | Yes                              |  |
|  | Downloading music                                                                         | yes                                   | No                               |  |

Since our dataset is basically Primary Data, a realistic assumption can be inferred very easily. out of total **315 data points**; **152 are female** variables , **163 are male** variables.

**A small-scale idea about the variables are given below:**

- 1. What is yourAge? :** Age of a person can also influence their music preference.
- 2. What is your MOTHER TONGUE?:** It mainly represents , in which language they speak.
- 3. What is your Gender? :** To understand more acutely about the music listening habits and its connection with psychological well-being , we divide the data into male and female.
- 4. What is your Zodiac/Star Sign?:** It's gives us a knowledge about our zodiac sign we mainly has 12 zodiac sign like Aries, Taurus, Gemini,Cancer, Leo, Virgo,Libra, Scorpio,Sagittarius ,Capricorn ,Aquarius ,Pisces
- 5. Do you listen to music while working?:**It's a yes/No type question .
- 6. Are you a MUSICIAN? :** It's a yes/No type question .
- 7. In which language do you prefer to listen to music? :** It gives us their language preference for listening to music.

8. **How many hours in a day do you listen to music?** : It gives us the house they spend while listening to music . It has four options: “less than 1 hour”, “1 hour – 3 hour” , “3 hour – 5 hour” , “More than 5 hour” .
9. **Mostly in which time do you prefer to listen to music? [Rock]** : It gives us , at what time in a day people prefer to listen to rock music.
10. **Mostly in which time do you prefer to listen to music? [Classical]** : It gives us , at what time in a day people prefer to listen to Classical music.
11. **Mostly in which time do you prefer to listen to music? [Slow + Reverb (Lofi)]** : It gives us , at what time in a day, people prefer to listen to Slow + Reverb (Lofi) music.
12. **Mostly in which time do you prefer to listen to music?[Instrumental]** : It gives us , at what time in a day, people prefer to listen to instrumental music.
13. **Mostly in which time do you prefer to listen to music? [Hip-Hop]**:It gives us , at what time in a day, people prefer to listen to hip-hop music.
14. **Mostly in which time do you prefer to listen to music? (one response in each column required) [Folk]** :It gives us , at what time in a day, people prefer to listen to folk music.

- 15. Mostly in which time do you prefer to listen to music?**  
[Romantic ]:It gives us , at what time in a day, people prefer to listen to devotional music.
- 16. How often do you listen to music because of the following reasons? [For entertainment]:** It gives us how frequently we listen to music for entertainment purposes. It's actually a 5 point likert scale.
- 17. How often do you listen to music because of the following reasons? [It evokes memories] :**It gives us how frequently we listen to music because it evokes memories. It's actually a 5 point likert scale.
- 18. How often do you listen to music because of the following reasons? [Interest in music itself]:**It gives us how frequently we listen to music because we have interest in music itself.It's actually a 5 point likert scale.
- 19. How often do you listen to music because of the following reasons? [Enjoy to listen to lyrics]:** It gives us how frequently we listen to music because we enjoy listening to lyrics.It's actually a 5 point likert scale.
- 20. How often do you listen to music because of the following reasons? [To enhance positive moods]:** It gives us how frequently we listen to music because it enhances positive mood.It's actually a 5 point likert scale.
- 21. How often do you listen to music because of the following reasons? [ To ease pressure]:** It gives us how frequently we

listen to music because it eases pressure .It's actually a 5 point likert scale.

22. **How often do you listen to music because of the following reasons? [To lessen your boredom]:** It gives us how frequently we listen to music because it helps to lessen our boredom .It's actually a 5 point likert scale.
23. **How common is it that you feel each of the following emotions in response to music? [Happy]:** It gives us how frequently we feel happy while listening to music. It's actually a 4 point likert scale.
24. **How common is it that you feel each of the following emotions in response to music? [Enjoying]:**It gives us how frequently we feel enjoying while listening to music. It's actually a 4 point likert scale.
25. **How common is it that you feel each of the following emotions in response to music? [Loving]:**It gives us how frequently we feel loving while listening to music. It's actually a 4 point likert scale.
26. **How common is it that you feel each of the following emotions in response to music? [Enchanted]:**It gives us how frequently we feel enchanted while listening to music. It's actually a 4 point likert scale.
27. **How common is it that you feel each of the following emotions in response to music? [Sad]:**It gives us how

frequently we feel sad while listening to music. It's actually a 4 point likert scale.

28. **Some aspects of your personality [Self Loving]:** It's a Yes/No type question. It gives us the person is self loving or not.
29. **Some aspects of your personality [Daring]:** It's a Yes/No type question. It gives us the person is daring or not.
30. **Some aspects of your personality [Introvert]:** It's a Yes/No type question. It gives us the person is introvert or not.
31. **Some aspects of your personality [Emotional]:** It's a Yes/No type question. It gives us the person is emotional or not.
32. **Some aspects of your personality [Optimistic ]:** It's a Yes/No type question. It gives us the person is optimistic or not.
33. **Which app do you use the most for listening to music ? [1st]**  
: It gives us our first preferable music listening app among "Youtube" , "Gaana" , "Spotify" , " Jiosaavn" and "wynk music".
34. **Which app do you use the most for listening to music ? [2nd]**  
: It gives us our second preferable music listening app among "Youtube" , "Gaana" , "Spotify" , " Jiosaavn" and "wynk music".
35. **Which app do you use the most for listening to music ? [3rd]**  
: It gives us our third preferable music listening app among "Youtube" , "Gaana" , "Spotify" , " Jiosaavn" and "wynk music".

36. When you don't have an internet connection then in what way do you get indulged in music?: It gives us without the internet how we enjoy music.
37. Do you listen to music while working? : It's a yes/No type question .
38. Does music have any effect or not? :It's a yes/No type question .

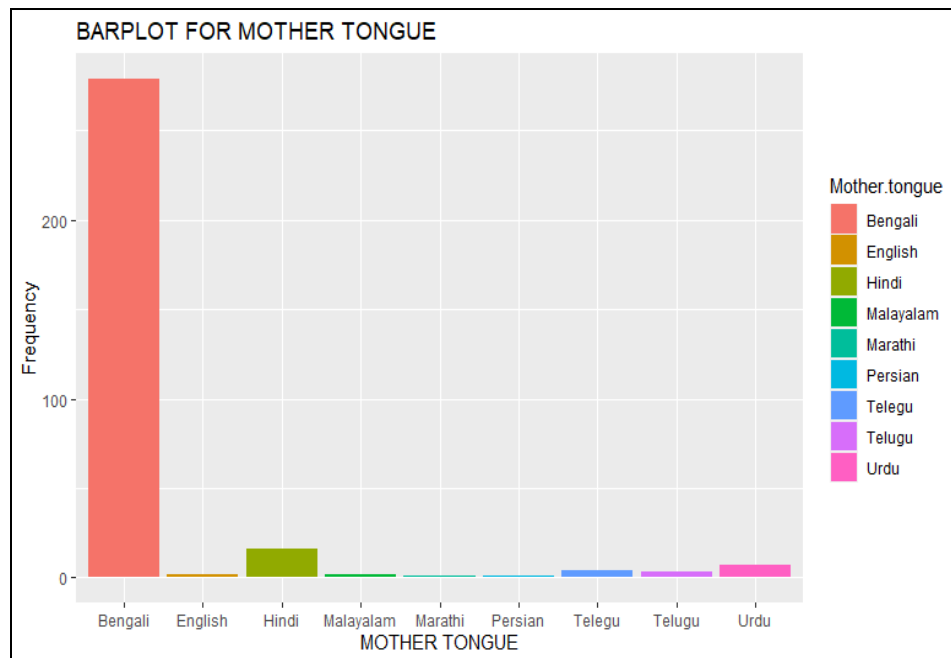
### 3. METHODOLOGY

In this project,R software has been used for the analysis. For the authenticity, a google form containing some relatable questions has been created (link in the Appendix).

### 3.1. Explanatory Data Analysis

#### 3.1.1 Diagrammatic representation

Diagrammatic Presentation of Data gives an immediate understanding of the real situation to be defined. Using the “ggplot” command, some divided bar diagrams along with some pie charts have been presented and just at a glance, an inference about Male, Female and finally the whole dataset can be drawn.

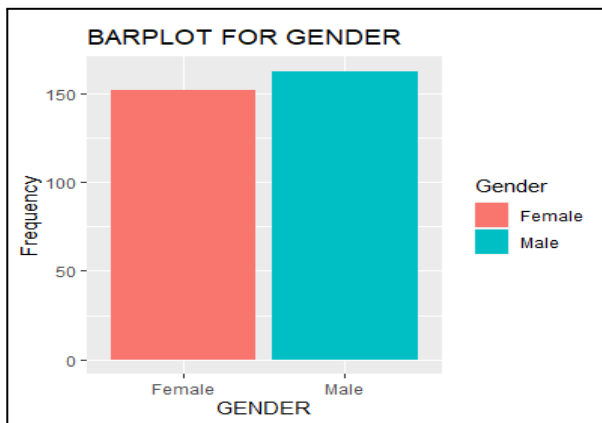
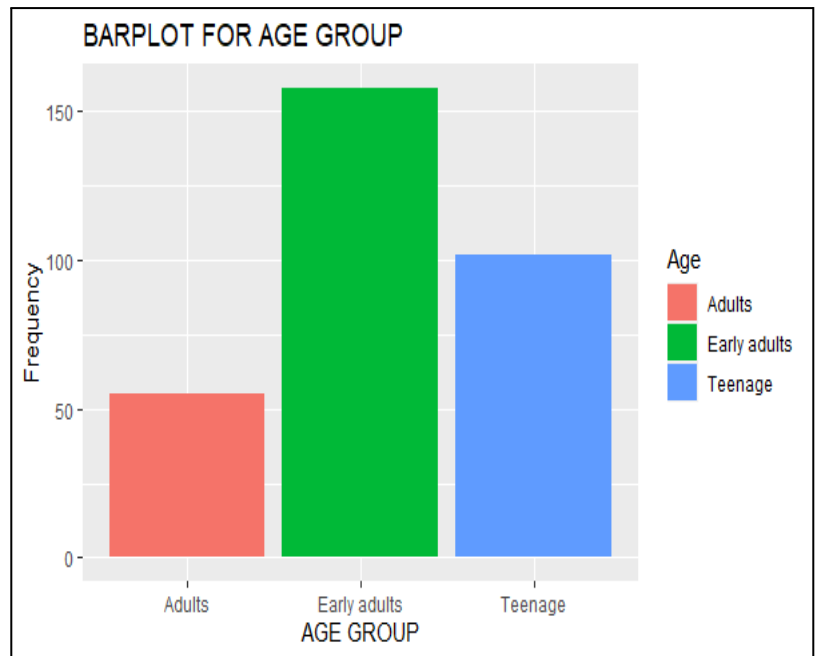


**Figure 1:** Mother tongue of observed people

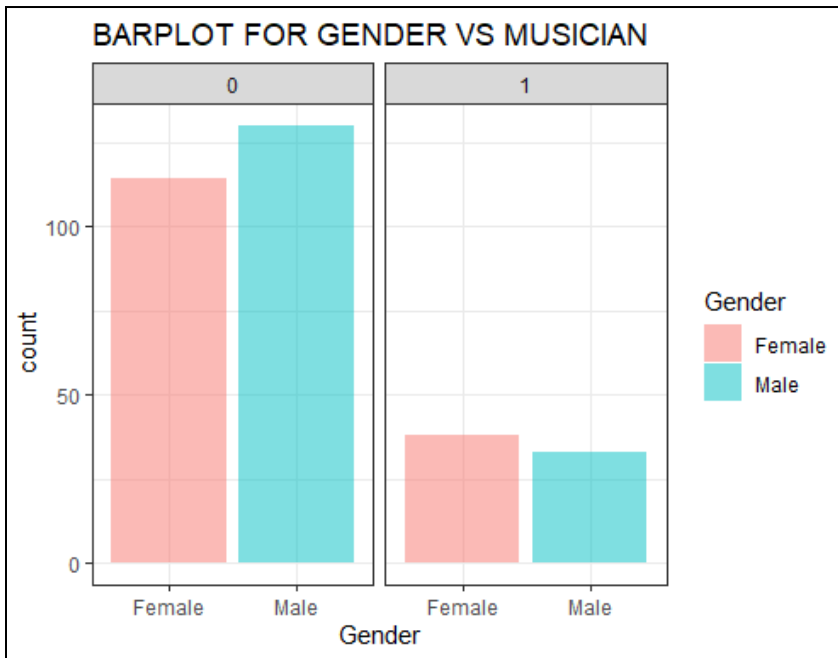
**Interpretation:** The data we get from our survey above 200 people speak in bengali also around 15 people speak in hindi and the number of people speak in any language other than bengali and hindi is very less

**Figure 2: Age of the observed data**

**Interpretation:** From the diagram as we can see that we get data around 53 of adults, more than 150 data of early adults and around 100 data on adults.



**Figure 3: Gender of the observed data**  
**Interpretation :** From the figure we can clearly see that male and female ratio is perfect in our dataset

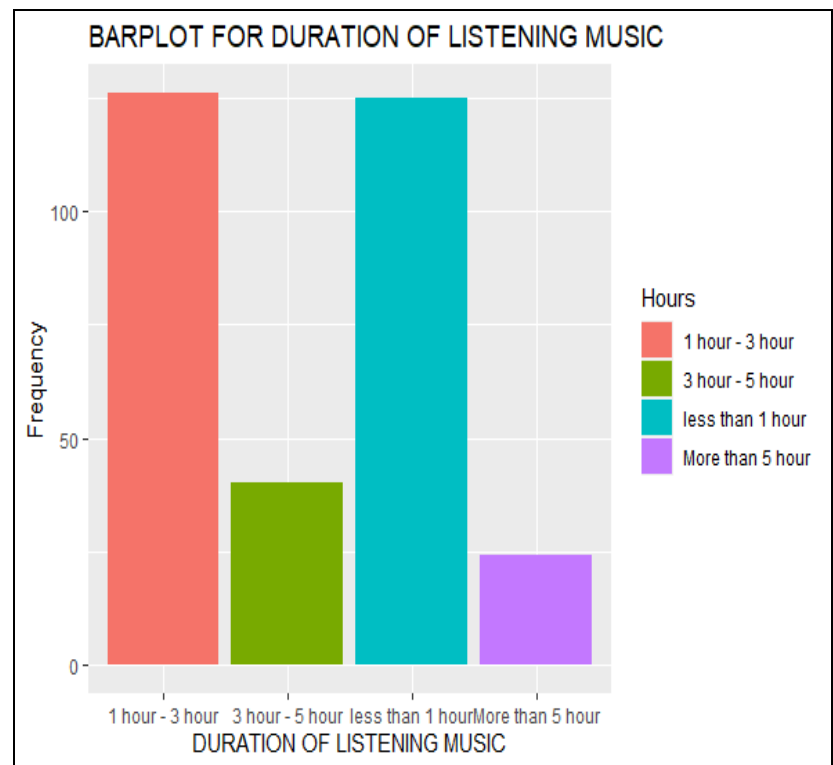


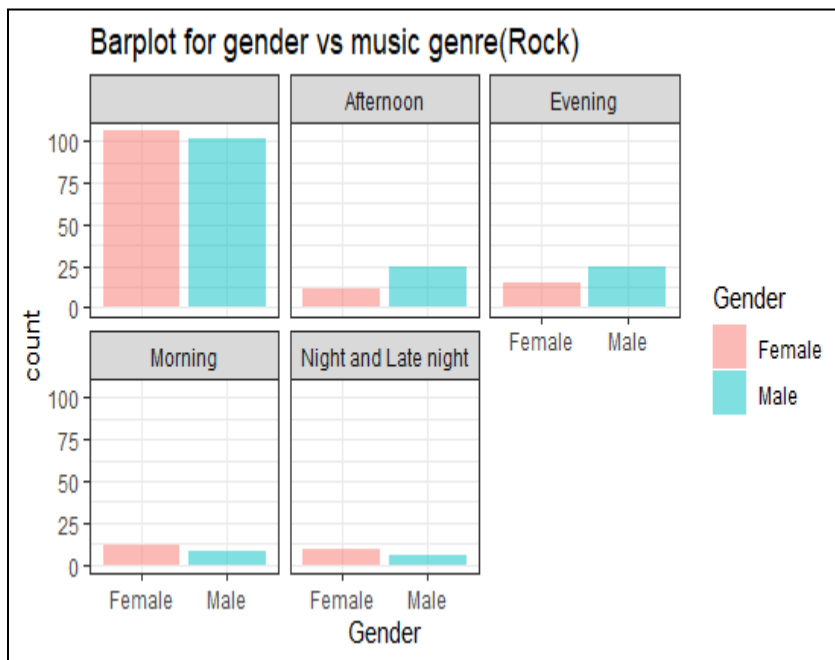
**Figure 4: gender vs musician**

**Interpretation :** From the figure we can see that number of female being musician is quite significant. Also we can see that number of male not being musician is also quite significant

**Figure 5: Duration of listening music**

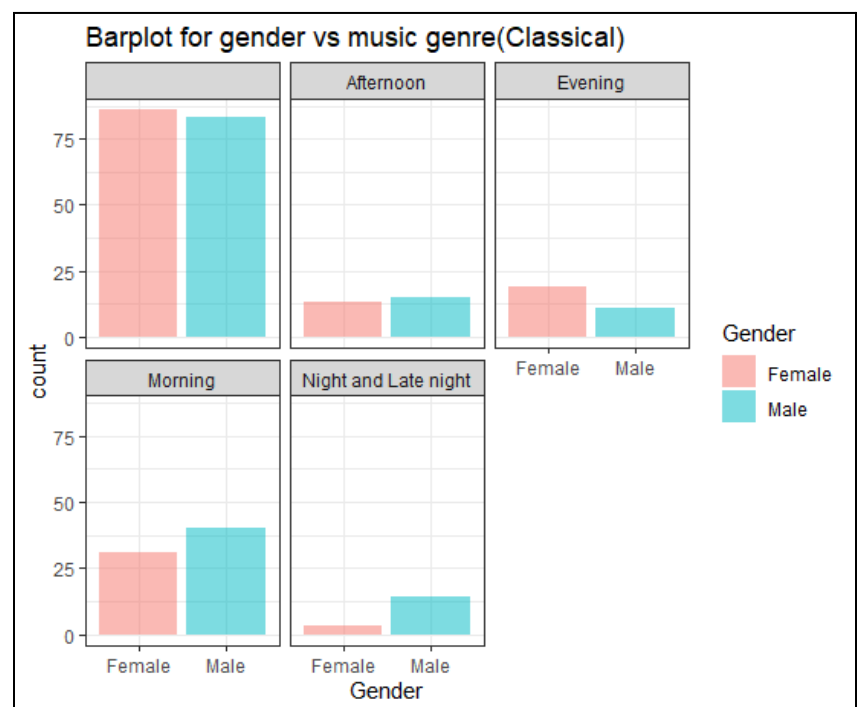
**Interpretation:** Here we can see that most people prefer to listen to music either for 1 hour to 3 hours or less than 1 hour. Very less people prefer to listen to music for more than 5 hours.

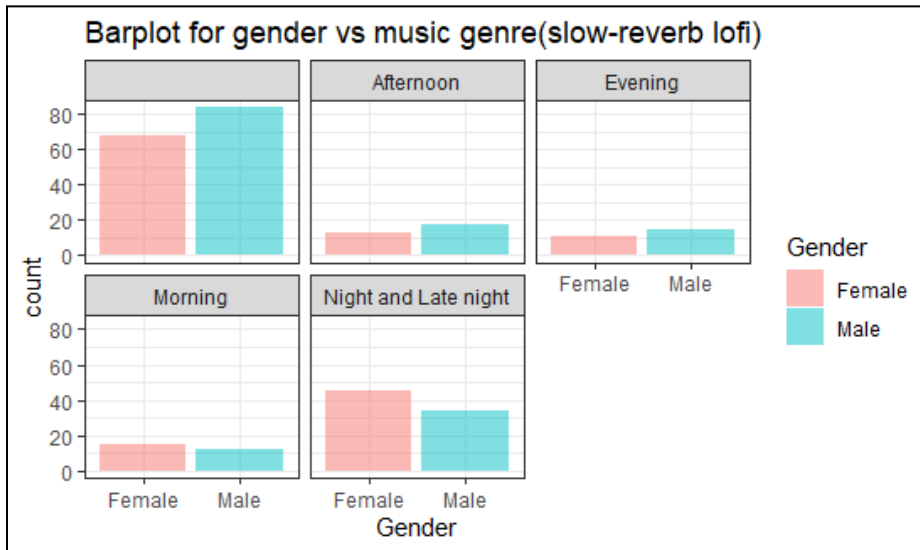




**Figure 6: Gender vs music genres(Rock)**  
**Interpretation:** we can see that the number of females not preferring rock music is significant. Also, the number of males preferring rock music in the afternoon and evening is quite high.

**Figure 7: Gender vs music genres(Classical)**  
**Interpretation:** we can see that the number of females not preferring classical music is significant. Also, the number of males preferring classical music in the morning and night and late night is more than the number of females. Also females preferred to listen to classical music in the afternoon and evening.





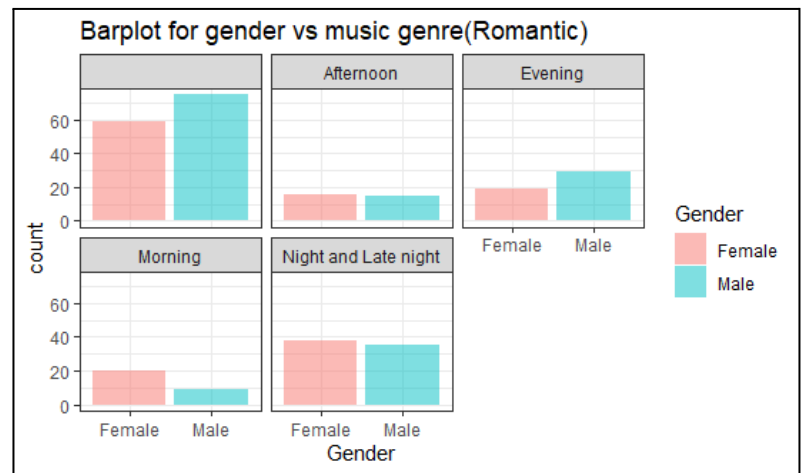
**Figure 8: Gender vs music genres (soft reverb lofi)**

**Interpretation:** we can see that the number of males not preferring soft reverb lofi music is significant. Also, the number of females preferring

classical music in the morning and night and late night is more than the number of males. Also males preferred to listen to classical music in the afternoon and evening.

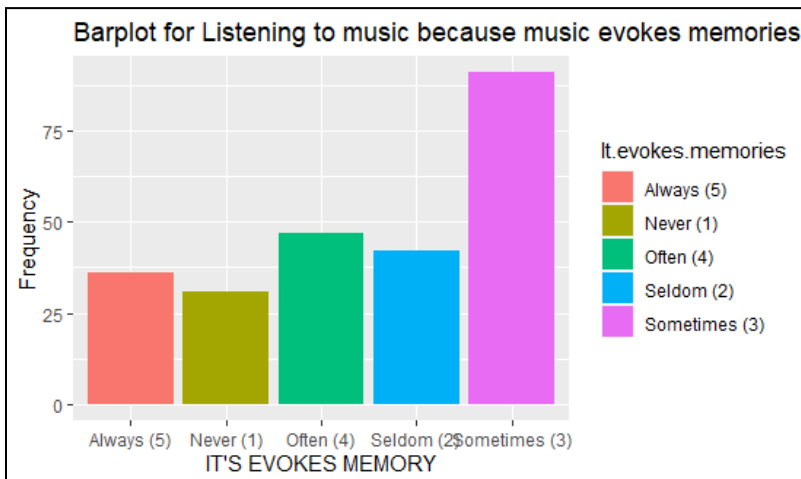
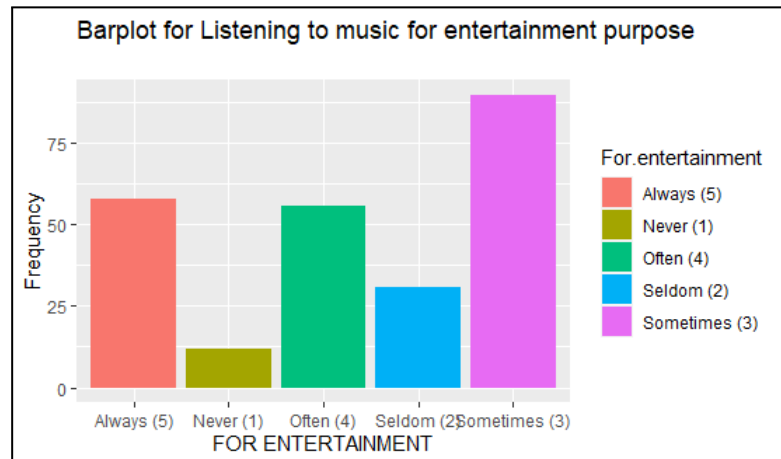
**Figure 9: Gender vs music genres (Romantic)**

**Interpretation:** we can see that the number of males not preferring romantic music is significant. Also, the number of males preferring romantic music in the afternoon morning and night and late night is more than the number of females. Also females preferred to listen to classical music in the morning, night and late night and evening.



**Figure 10:** Listening to music for entertainment purpose

**Interpretation:** we can see that most people use music for their entertainment purposes.

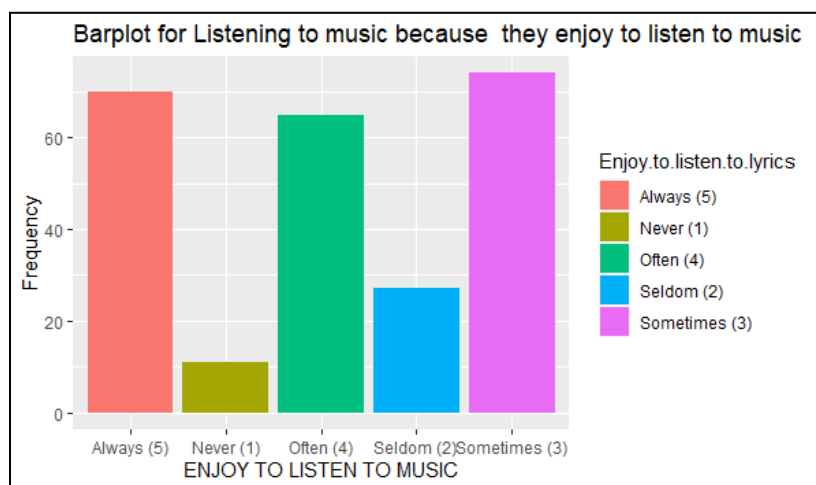


**Figure 11:** Listening to music because music evokes memories

**Interpretation:** we can see that most people sometimes use music for because music evokes memories

**Figure 12:** Listening to music because they enjoy to listen to music

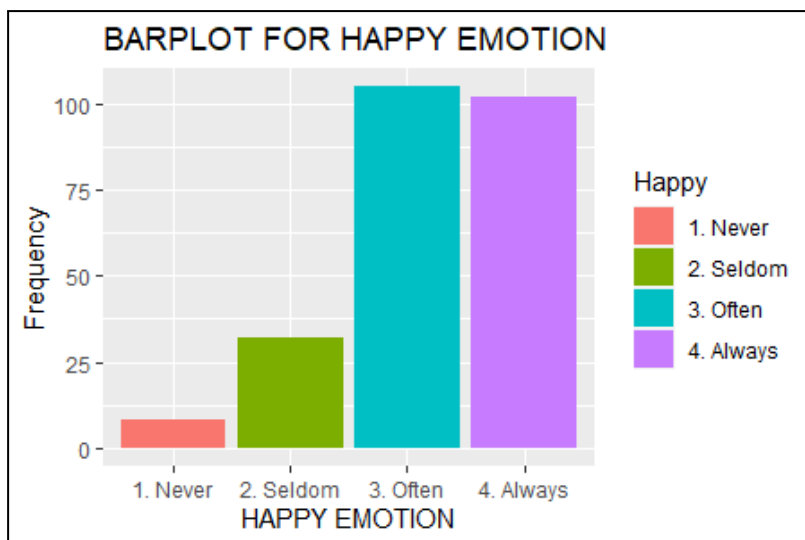
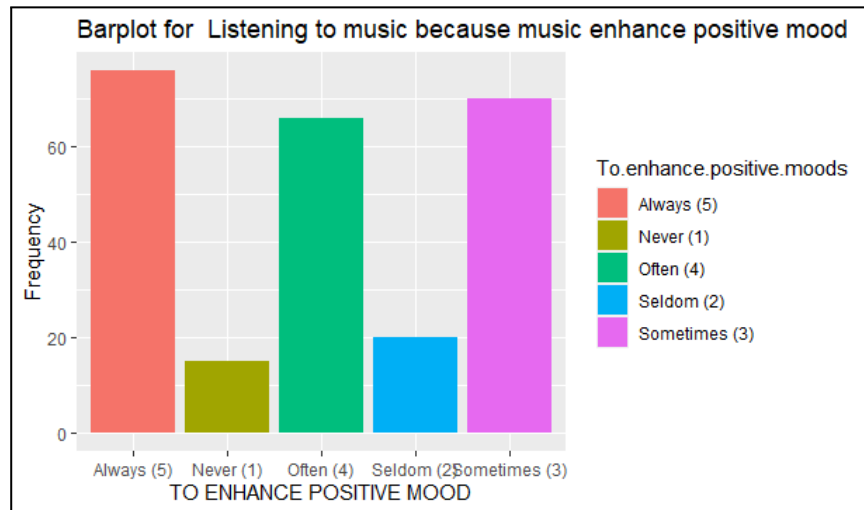
**Interpretation:** Here we can see that people always often and sometimes listen to music because they enjoy to listen to music



**Figure 13:**  
Listening to music  
because music  
enhance positive  
mood

**Interpretation:**

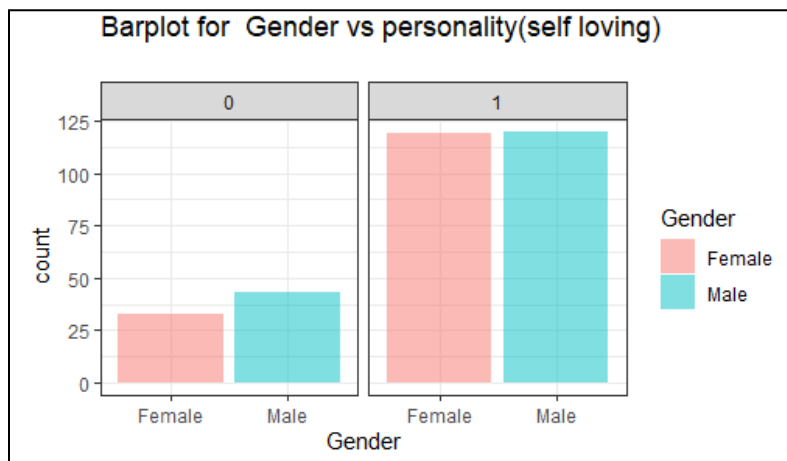
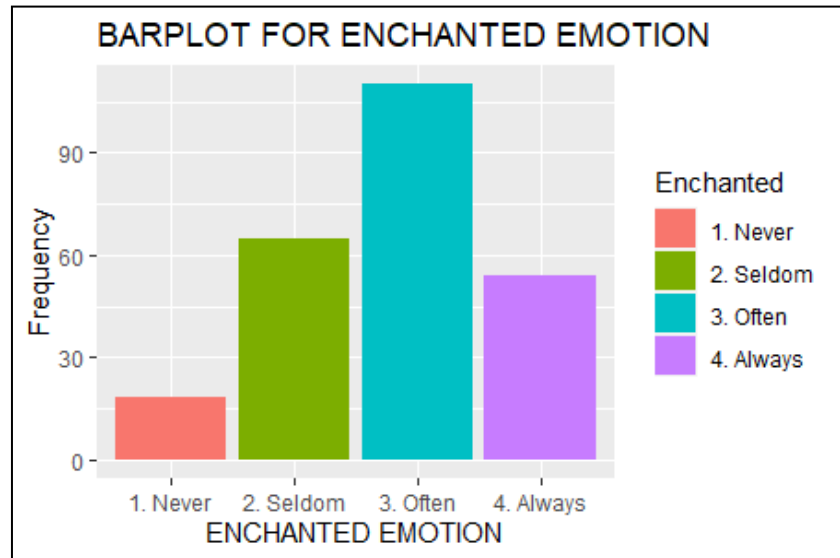
Here we can see  
that people always  
listen to music  
because music  
enhances a  
positive mood.



**Figure 14:** Feeling  
happy after listening  
to music

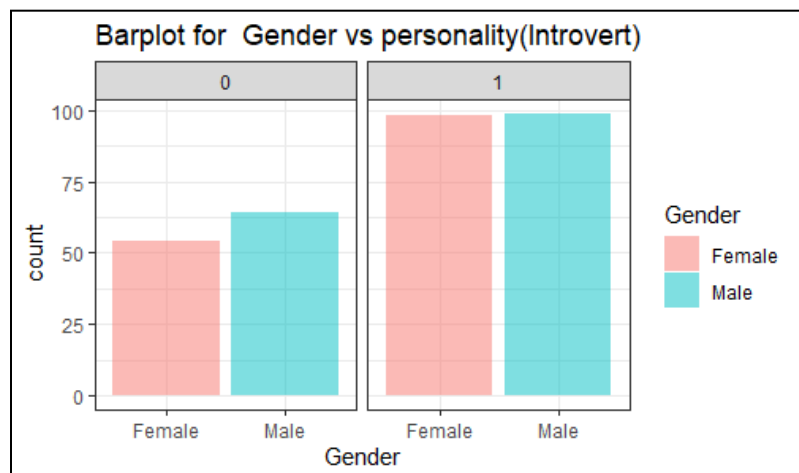
**Interpretation:** We  
can see that people  
mostly respond that  
they often and  
always feel happy  
after listening to  
music.

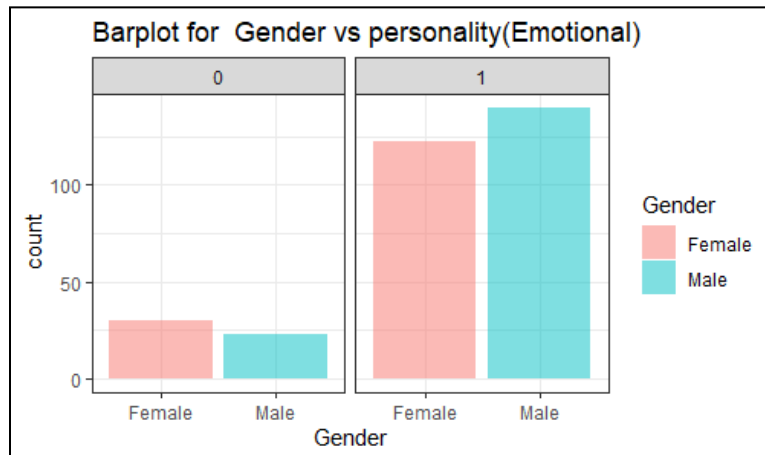
**Figure 15:** Feeling enchanted after listening to music  
**Interpretation:** We can see that people mostly respond that they seldom often and always feel enchanted after listening to music.



**Figure 16:** Gender vs personality(self loving)  
**Interpretation:** we can see that the number of females not being selfloving is significant.

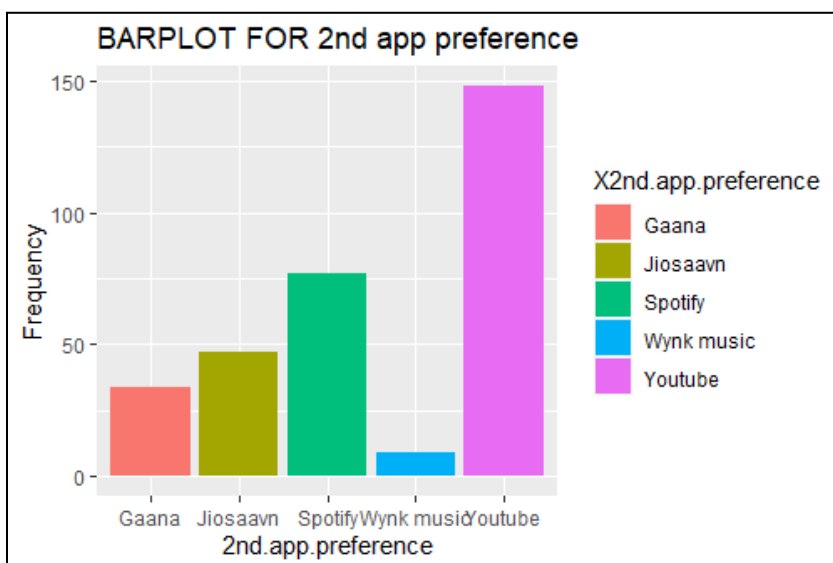
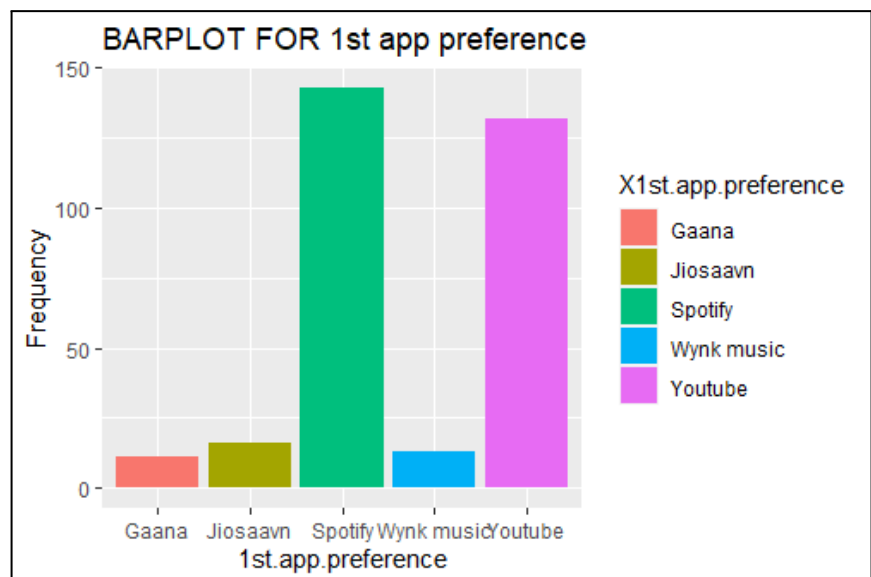
**Figure 17:** Gender vs personality(introvert)  
**Interpretation:** we can see that the number of males being introvert is significant





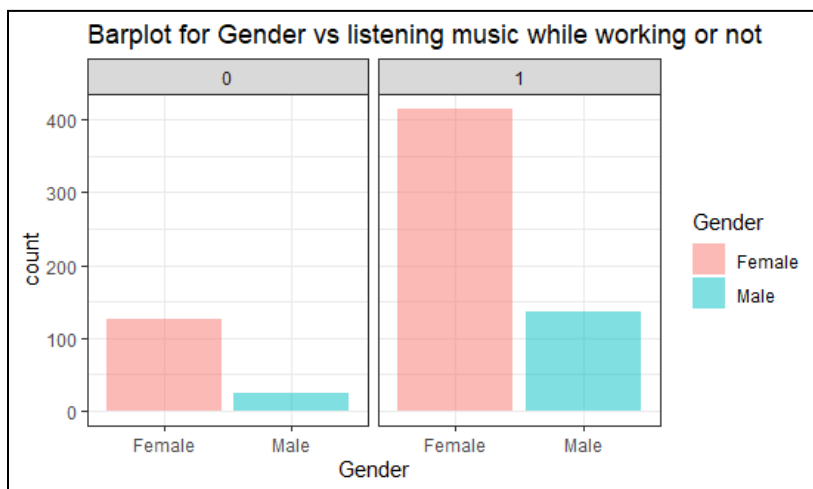
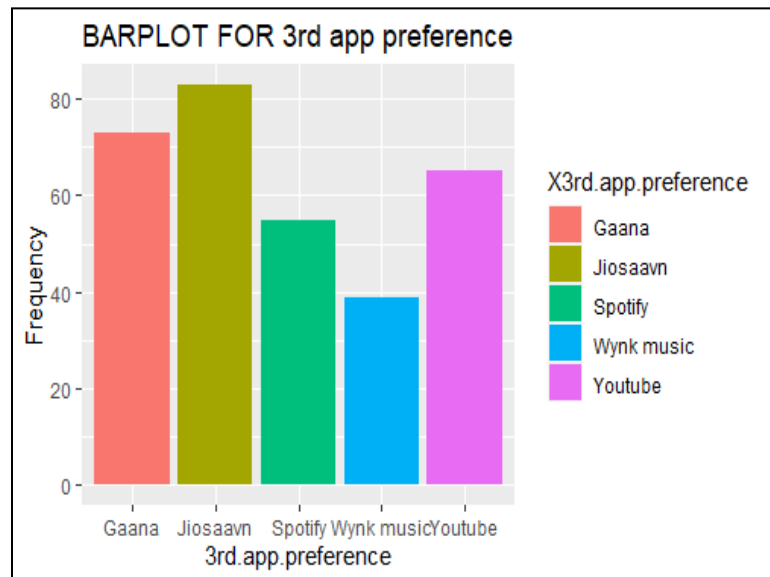
**Figure 18: Gender vs personality(emotional)**  
**Interpretation:** we can see that the number of males being emotional is significant.

**Figure 19: 1st preferable music listening app**  
**Interpretation:** We can see that people choose spotify music listening apps as their 1st preference.



**Figure 20: 2nd preferable music listening app**  
**Interpretation:** We can see that people choose youtube music listening apps as their 2nd preference.

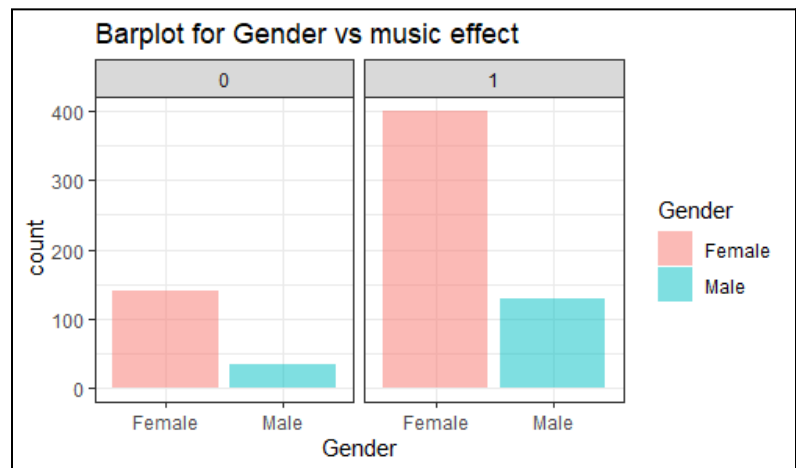
**Figure 21:** 3rd preferable music listening app  
**Interpretation:** We can see that people choose jiosaavn music listening apps as their 3rd preference.



**Figure 22:** Gender vs listening music while working or not  
**Interpretation:** Number of female listen to music is significant as well as number of male not listening music while working is quite less

**Figure 23: Gender vs music effect**

**Interpretation:** Number of female listen to music is significant as well as number of male not listening music while working is quite less



### 3.2 Chi square test of association

The Chi-square Test provides a method for testing the association between the row and column variables in a two-way table (contingency table) consisting of categorical data. The null hypothesis  $H_0$  assumes that there is no association between the variables (in other words, one variable does not vary according to the other variable), while the alternative hypothesis  $H_a$  claims that some association does exist.

#### 3.2.1 Test statistic

This test is based on a test statistic that measures the divergence of the observed data from the values that would be expected under the null hypothesis of no association. It is also known as Pearsonian  $\chi^2$  statistic. It is given as,

$\chi^2 = [(\sum_{i=1}^r \sum_{j=1}^s f_{ij} - np_{ij})^2] / np_{ij}$  can be used and

$\chi^2 \sim \chi^2_{rs-1}$  (asymptotically), the only relation being  $\sum_{i=1}^r \sum_{j=1}^s f_{ij} = n$  Under  $H_0$ , we have  $p_{ij} = p_{i0} p_{0j}$ ,  $\forall (i,j)$  and the statistic reduces to,

$\chi^2 = [(\sum_{i=1}^r \sum_{j=1}^s f_{ij} - np_{i0} p_{0j})^2] / np_{i0} p_{0j}$ . Here,  $p_{i0}$ 's and  $p_{0j}$ 's are unknown and are to be estimated from the sample. There are exactly  $(r+s)$

parameters  $p_{i0}$  and  $p_{0j}$ , with  $\sum_{i=1}^r p_{i0} = \sum_{j=1}^s p_{0j} = 1$ ;

i.e.,  $(r+s-2)$  independent parameters which are to be obtained from the sample.

By Maximum Likelihood Method, we get  $\widehat{p_{i0}} = \frac{f_{i0}}{n}$  and  $\widehat{p_{0j}} = \frac{f_{0j}}{n}$ . Hence, under  $H_0$ ,

$$\begin{aligned} \chi^2 &= [(\sum_{i=1}^r \sum_{j=1}^s f_{ij} - np_{i0} p_{0j})^2] / np_{i0} p_{0j} \\ &= n \left\{ \sum_i \sum_j \frac{f_{ij}^2}{f_{i0} f_{0j}} - 1 \right\} \end{aligned}$$

tend a  $\chi^2$  distribution with  $(rs-1)-(r+s-2)=(r-1)(s-1)$  d.f., since  $(r+s-2)$  independent parameters have been estimated,  $n \rightarrow \infty$ . Here, the statistic  $\chi^2$  measures the departure from independence. Hence, we reject  $H_0$  at level  $\alpha$ , if  $\chi^2_{obs} \geq \chi^2_{\alpha; (r-1)(s-1)}$ , provided  $n$  is large enough. We Reject the null hypothesis at level 0.05 if,  $p\text{-value} = P(\chi^2 \geq \chi^2_{obs}) \leq 0.05$ .

### 3.2.2 Interpretation Of Our Data Based On Test

**Table :2**  
**contingency table of Music effect on musician**

| <b>Musician</b> | <b>Music effect</b> |            | <b>Total</b> |
|-----------------|---------------------|------------|--------------|
|                 | <b>No</b>           | <b>Yes</b> |              |
| <b>No</b>       | 80                  | 70         | 150          |
| <b>Yes</b>      | 75                  | 85         | 155          |
| <b>Total</b>    | 150                 | 155        | 305          |

$$\chi^2 = 1.723$$

$$p\text{-value} = 0.1893$$

**Interpretation:** This Table depicts how Music effects are not associated with the status of a musician. Here,  $p\text{-value} > 0.05$ , thus the null hypothesis is Accepted.

**Table : 3**  
**contingency table of Music effect and classical music listening people**

| <b>Classical</b> | <b>Music effects</b> |            | <b>Total</b> |
|------------------|----------------------|------------|--------------|
|                  | <b>No</b>            | <b>Yes</b> |              |
| <b>No</b>        | 64                   | 87         | 151          |
| <b>Yes</b>       | 86                   | 68         | 154          |
| <b>Total</b>     | 150                  | 155        | 305          |

$$\chi^2 = 5.0013$$

$$df = 1$$

$$p\text{-value} = 0.02533$$

**Interpretation :** This Table depicts how Music effects are associated with the status of Classical listeners in working hours. Here,  $p\text{-value} < 0.05$ , thus the null hypothesis is Rejected

**Table:4**  
**contingency table of Music effect and Frequency of while working**

| While working | Music effects |     | Total |
|---------------|---------------|-----|-------|
|               | No            | Yes |       |
| No            | 80            | 71  | 151   |
| Yes           | 70            | 84  | 154   |
| Total         | 150           | 155 | 305   |

$$\chi^2 = 1.4397$$

$$df = 1$$

$$p\text{-value} = 0.2302$$

**Interpretation:** This Table depicts how Music effects are not associated with the status of listeners in working hours. Here,  $p\text{-value} > 0.05$ , thus the null hypothesis is Accepted.

**Table:5**

**Contingency table for age and music effect**

| Age group          | Music effects |     | Total |
|--------------------|---------------|-----|-------|
|                    | No            | Yes |       |
| 10-30 years        | 38            | 39  | 77    |
| 30-51 years        | 78            | 66  | 144   |
| More than 51 years | 34            | 50  | 84    |
| Total              | 150           | 155 | 305   |

$$\chi^2 = 3.9797$$

$$df = 2$$

$$p\text{-value} = 0.1367$$

**Interpretation:** This Table depicts how Music\_effects\_ is not associated with the different age groups. Here,  $p\text{-value} > 0.05$ , thus the null hypothesis is Accepted.

Therefore, summarizing all the results above, we note that a chi-square test for association has been performed for checking the dependency of one's opinion about effects of music in their working hours on variates such as their Age, Passion for music, listening classical while working, listens classical song as their preference .

Surprisingly, we have found out that the opinions are dependent on only one criterion i.e, "listens classical song as their preference" and all other covariates i.e Age, Passion for music, listening classical while working do not affect their opinion.

**Table:6**

**Contingency table for music genres and time in a day  
(Morning,afternoon,evening,night and late night)**

| Time in a day        | Music genres |               |             |            |             |      | Total |
|----------------------|--------------|---------------|-------------|------------|-------------|------|-------|
|                      | Classical    | Instrume-ntal | Slow reverb | Romant-i c | Devotion al | folk |       |
| Morning              | 11           | 11            | 13          | 7          | 6           | 14   | 62    |
| Afternoon            | 16           | 6             | 18          | 16         | 10          | 2    | 68    |
| Evening              | 15           | 10            | 10          | 20         | 8           | 5    | 68    |
| Night and late night | 39           | 22            | 8           | 10         | 17          | 3    | 99    |
| <b>Total</b>         | 81           | 49            | 49          | 53         | 41          | 24   | 297   |

$$\chi^2 = 56.739$$

$$df = 15$$

$$p\text{-value} = 9.087e-07$$

**Interpretation:-** This Table depicts how Music genre preference is associated with the different time points in a day. Here,  $p\text{-value} < 0.05$ , thus the null hypothesis is Rejected, i.e there is sufficient evidence of association between Music genre preference & the different time points in a day.

### 3.2.3 Inference

A surprising part of the result is that most of the pairs of variables may be significantly associated in the total population, but independent in the gender based sub-populations and vice-versa. In our study, this eventually introduces Simpson's Paradox, which is a

statistical phenomenon where an association between two variables in a population emerges, disappears or reverses when the population is divided into sub-populations.

### **3.3 Analysis of variance**

#### **3.3.1 Definition**

ANOVA, which stands for Analysis of Variance, is a statistical test used to analyze the difference between the means of more than two groups.

A one-way ANOVA uses one independent variable, while a two-way ANOVA uses two independent variables.

#### **3.3.2 Use a one-way ANOVA**

when you have collected data about one categorical independent variable and one quantitative dependent variable. The independent variable should have at least three levels (i.e. at least three different groups or categories). ANOVA tells you if the dependent variable changes according to the level of the independent variable.

#### **3.3.3 Assumptions of ANOVA**

For the results of a one-way ANOVA to be valid, the following assumptions should be met:

1. **Normality** – Each sample was drawn from a normally distributed population.

2. **Equal Variances** – The variances of the populations that the samples come from are equal. You can use Bartlett's Test to verify this assumption.

3. **Independence** – The observations in each group are independent of each other and the observations within groups were obtained by a random sample.

A one-way ANOVA uses the following null and alternative hypotheses:

$H_0$  (**null hypothesis**):  $\mu_1 = \mu_2 = \mu_3 = \dots = \mu_K$  (all the population means are equal)

VS

$H_1$  (**alternative hypothesis**): at least one population mean is different from the rest .

No matter which software you use, you will receive the following table as output:

**Table:7**  
**Anova table**

| S.OV      | df     | SS  | MS  | F                 | p                |
|-----------|--------|-----|-----|-------------------|------------------|
| Treatment | $df_r$ | SSR | MSR | $\frac{MSR}{MSE}$ | $F_{df_r, df_e}$ |
| Error     | $df_e$ | SSE | MSE |                   |                  |
| Total     | $df_t$ | SST |     |                   |                  |

k: total number of groups

n: total observations

SSR: regression sum of squares

SSE: error sum of squares

SST: total sum of squares (SST = SSR + SSE)

$df_r$ : regression degrees of freedom ( $df_r = k-1$ )

$df_e$ : error degrees of freedom ( $df_e = n-k$ )

$df_t$ : total degrees of freedom ( $df_t = n-1$ )

MSR: regression mean square ( $MSR = SSR/df_r$ )

MSE: error mean square ( $MSE = SSE/df_e$ )

F: The F test statistic ( $F = MSR/MSE$ )

p: The p-value that corresponds to  $F_{df_r, df_e}$

If the p-value is less than your chosen significance level (e.g. 0.05), then you can reject the null hypothesis and conclude that at least one of the population means is different from the others.

**Note:** If you reject the null hypothesis, this indicates that at least one of the population means is different from the others, but the ANOVA table doesn't specify which population means are different. To determine this, you need to perform post hoc tests, also known as "multiple comparisons" tests.

### 3.3.4 Tests based on 1-way ANOVA

#### [Test-1]

Dividing the "Age" column into three groups

- Group1: 10-30 years
- Group2: 31-51 years
- Group3: Above 52 years

[ Lack of sufficient data in the age group:73- 93 years, we have merged the age group "52-72 years" and "73-93 years" as "Above 52 years"]

Each group has entries of music consumption time for each individual(in hours). So we want to test whether there is any effect of

age grouping on music listening time( $H_a$ ) or not( $H_0$  ). The test report is as follows:

**Table:8**  
**Anova: Single Factor (summary)**

| Group  | Count | Sum   | Average  | Variance |
|--------|-------|-------|----------|----------|
| Group1 | 256   | 543.6 | 2.123438 | 5.168939 |
| Group2 | 16    | 18.2  | 1.1375   | 0.553167 |
| Group3 | 24    | 29.8  | 1.241667 | 0.940797 |

**Table:9**  
**Anova table for Single Factor**

| S.O.V          | SS       | df  | MS       | F       | P-value  | F crit   |
|----------------|----------|-----|----------|---------|----------|----------|
| Between groups | 29.60425 | 2   | 14.80213 | 3.21734 | 0.041481 | 3.026572 |
| Within groups  | 1348.015 | 293 | 4.600734 |         |          |          |
| Total          | 1377.619 | 295 |          |         |          |          |

### 3.3.5 Interpretation

Based on the provided table, the p-value for the "Between Groups" variation is less than the significance level (typically 0.05). Therefore, we can reject the null hypothesis and conclude that there is sufficient

evidence to suggest a significant difference between at least two of the age groups.

Now, to test which groups have significant differences between them, let us perform a pairwise t-test for each of two age groups.

**Table:10**

**t-Test: Two-Sample Assuming Unequal Variances(Group1,Group2)**

|                                     | <b>Group1</b> | <b>Group2</b> |
|-------------------------------------|---------------|---------------|
| <b>Mean</b>                         | 2.123438      | 1.1375        |
| <b>Variance</b>                     | 5.168939      | 0.553167      |
| <b>observation</b>                  | 256           | 16            |
| <b>Hypothesized mean difference</b> | 0             |               |
| <b>df</b>                           | 37            |               |
| <b>t stat</b>                       | 4.213097      |               |
| <b>P(T&lt;=t)one tail</b>           | 7.75E-05      |               |
| <b>t critical one tail</b>          | 1.687094      |               |
| <b>P(T&lt;=t)two tail</b>           | 0.000155      |               |
| <b>t critical two tail</b>          | 2.026192      |               |

**Interpretation:-** Based on the provided results, the t-test indicates a significant difference between the two groups. The t Stat value of 4.213097144 is greater than the critical t value (either one-tail or two-tail). The p-values (both one-tail and two-tail) are less than the significance level (typically 0.05). Therefore, we can reject the null hypothesis and conclude that there is a significant difference between the means of Group1 and Group2

**Table:11**

**t-Test: Two-Sample Assuming Unequal Variances(Group1,Group3)**

|                                     | <b>Group1</b> | <b>Group3</b> |
|-------------------------------------|---------------|---------------|
| <b>Mean</b>                         | 2.123438      | 1.241667      |
| <b>Variance</b>                     | 5.168939      | 0.940797      |
| <b>observation</b>                  | 256           | 24            |
| <b>Hypothesized mean difference</b> | 0             |               |
| <b>df</b>                           | 52            |               |
| <b>t stat</b>                       | 3.618222      |               |
| <b>P(T&lt;=t)one tail</b>           | 0.000336      |               |
| <b>t critical one tail</b>          | 1.674689      |               |
| <b>P(T&lt;=t)two tail</b>           | 0.000671      |               |
| <b>t critical two tail</b>          | 2.006647      |               |

**Interpretation:-** Based on the provided results, the t-test indicates a significant difference between the two groups. The t Stat value of 3.618222266 is greater than the critical t value (either one-tail or two-tail). The p-values (both one-tail and two-tail) are less than the significance level (typically 0.05). Therefore, we can reject the null hypothesis and conclude that there is a significant difference between the means of Group1 and Group3.

**Table:12**

**t-Test: Two-Sample Assuming Unequal Variances(Group2,Group3)**

|                              | Group2   | Group3   |
|------------------------------|----------|----------|
| Mean                         | 1.1375   | 1.241667 |
| Variance                     | 0.553167 | 0.940797 |
| observation                  | 16       | 24       |
| Hypothesized mean difference | 0        |          |
| df                           | 37       |          |
| t stat                       | -0.38351 |          |
| P(T<=t)one tail              | 0.351767 |          |
| t critical one tail          | 1.687094 |          |
| P(T<=t)two tail              | 0.703534 |          |
| t critical two tail          | 2.026192 |          |

**Interpretation:-** Based on the provided results, the t-test does not indicate a significant difference between the two groups. The t Stat value of -0.383513491 is less than the critical t value (either one-tail or two-tail). The p-values (both one-tail and two-tail) are greater than the significance level (typically 0.05). Therefore, we fail to reject the null hypothesis and conclude that there is sufficient evidence to suggest a significant difference between the means of Group2 and Group3.

### 3.4 Multiple logistic regression

Multiple logistic regression is a statistical modeling technique used to predict the probability of a binary outcome (a dichotomous variable

with two categories, often coded as 0 and 1) based on multiple predictor variables. It is an extension of simple logistic regression, which considers only one predictor variable.

In multiple logistic regression, the goal is to find the relationship between the binary response variable and two or more predictor variables by estimating the coefficients of the predictors. These coefficients represent the change in the log-odds of the outcome associated with a one-unit change in the predictor variable while holding other variables constant.

The model assumes that the relationship between the predictor variables and the log-odds of the binary outcome is linear on the logit scale. The logit function is the natural logarithm of the odds, and it maps the probability of success (the probability of the binary outcome being 1) to a continuous scale.

### 3.4.1 Assumption

Multiple logistic regression, like other regression models, relies on several assumptions to provide reliable and valid results. Here are the key assumptions of multiple logistic regression:

**Binary outcome:** The dependent variable (response variable) should be binary or dichotomous, representing two categories or classes. Logistic regression is specifically designed for modeling binary outcomes.

**Linearity of predictors and log-odds:** The relationship between the predictors and the log-odds of the outcome should be linear. This assumption implies that the relationship between each predictor variable and the log-odds of the outcome is a straight line when plotted on a logit scale.

**Independence of observations:** Observations in the dataset should be independent of each other. Independence assumption ensures that

each observation provides unique information and that the model does not violate the assumption of independence in its estimation process.

**Absence of multicollinearity:** Predictor variables should be minimally correlated with each other. High correlations among predictors (multicollinearity) can lead to unstable estimates, make it difficult to interpret the individual effects of predictors, and affect the model's performance.

**Absence of influential outliers:** The presence of influential outliers can affect the estimated coefficients and undermine the validity of the model. It is important to check for and handle outliers appropriately.

**Adequate sample size:** Logistic regression typically requires a sufficient sample size to provide stable estimates and accurate inference. The rule of thumb is to have at least 10-15 observations per predictor variable to avoid overfitting and obtain reliable results.

**Assumption of no perfect separation:** Perfect separation occurs when the model can perfectly predict the outcome based on a combination of predictor variables, leading to infinite parameter estimates and unreliable inference. Techniques such as penalized regression or adding regularization terms can help address this issue. It is essential to assess these assumptions when applying multiple logistic regression and consider potential violations that may impact the interpretation and validity of the results. Diagnostic tools, such as residual analysis and goodness-of-fit tests, can help evaluate the model's assumptions.

### 3.4.2 Logit transformation

In multiple logistic regression, the relationship between the predictor variables and the log-odds of the binary outcome is assumed to be

linear on the logit scale. The logit transformation is used to convert the probability of success (the probability of the binary outcome being 1) to a continuous scale that spans from negative infinity to positive infinity.

The logit function is defined as the natural logarithm of the odds, which is the ratio of the probability of success (p) to the probability of failure (1 - p):

$$\text{logit}(p) = \log(p / (1 - p))$$

In the context of multiple logistic regression, the model can be mathematically represented as:

$$\text{logit}(p) = \beta_0 + \sum_{i=1}^n \beta_i X_i$$

where:

logit(p) is the log-odds of the probability p of the binary outcome being 1,

$\beta_0$  is the intercept or constant term,

$\beta_i$  's are the coefficients corresponding to the predictor variables

$X_i$  's respectively where  $i=1,2,\dots,n$

The logit transformation is useful in logistic regression because it allows us to model the linear relationship between the predictor variables and the log-odds of the binary outcome. This linear relationship simplifies the estimation process and facilitates the interpretation of the coefficients.

Once the logistic regression model is fitted, the predicted logit values can be converted back to probabilities using the inverse of the logit function, which is called the logistic function (also known as the sigmoid function):

$$p = 1 / (1 + \exp(-\text{logit}(p)))$$

where:

$p$  is the probability of success (the probability of the binary outcome being 1),

$\text{logit}(p)$  is the logit value obtained from the logistic regression model.

The logistic function maps the logit values to probabilities that range between 0 and 1, making it suitable for representing probabilities of binary outcomes. These probabilities can then be used to make predictions on new data or evaluate the performance of the logistic regression model.

### 3.4.3 General model

The form of the model is ,

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 * X_1 + ..... + \beta_k * X_k \quad (3.3)$$

In this model,  $p$  is the probability that the dependent variable  $Y=1$  and  $X_i$  's are the independent variables (predictors).  $\beta_0$  is a constant and  $\beta_i$  's are known as the regression coefficients, which have to be estimated from the data. Logistic regression estimates the probability of a certain event occurring.

Logistic regression thus forms a predictor variable ( $\log(p/(1-p))$ ) which is a linear combination of the explanatory variables. The values of this predictor variable are then transformed into probabilities by a logistic function. Such a function has the shape of an S called "Sigmoid curve".

### 3.4.4 Removing outliers, influential and leverages

In the project few influences and leverage are detected and finally removed. In logistic regression, a set of observations whose values deviate from the expected range and produce extremely large residuals and may indicate a sample peculiarity is called outliers. These outliers can unduly influence the results of the analysis and lead to incorrect

inferences. An observation is said to be influential if removing the observation substantially changes the estimate of coefficients. Influence can be thought of as the product of leverage and outliers. An observation with an extreme value on a predictor variable is called a point with high leverage. Leverage is a measure of how far an independent variable deviates from its mean. Detection of outliers and influential cases and corresponding treatment is a very crucial task of any modeling exercise. A failure to detect outliers and hence influential cases can have severe distortion on the validity of the inferences drawn from such modeling exercise

### 3.4.5 Interpretation

#### 3.4.5.1 Interpretation of the first model

We wish to fit our Logistic Model, for which a binary response is required, which corresponds to being a musician or not(Y). Here, our explanatory variables or covariates are the following:

- $X_1$ = Age
- $X_2$ =Mother tongue
- $X_3$ =Zodiac sign
- $X_4$ = Language preference

The regression model is given by,

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \sum_{i=1}^4 \beta_i X_i + e$$

The coefficient estimates, standard errors, z-values, and p-values are provided for each covariate. The p-values indicate the significance of each covariate in predicting the response variable. In this case, zodiac signs are significant at 0.05 level of significance. The other covariates (Age, Mother tongue and language preference is not) are not significant at 0.05 level of significance. The AIC (Akaike Information

Criterion) for this model is 352.4. A lower AIC value indicates a better-fitting model.

#### 3.4.5.2 Interpretation of the second model

We wish to fit our Logistic Model, for which a binary response is required, which corresponds to Devotional Music\_effects(Y). Here, our explanatory variables or covariates are the following:

- $X_1$ = Age
- $X_2$ =Listens music While working or not
- $X_3$ = During working hours, listens devotional music or no
- $X_4$ = Passionate for music or not

The regression model is given by,

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \sum_{i=1}^4 \beta_i X_i + e$$

The coefficient estimates, standard errors, z-values, and p-values are provided for each covariate. The p-values indicate the significance of each covariate in predicting the response variable. In this case, the covariate Frequency\_Romantic has a p-value of 0.0207, which suggests it is significant at the 0.05 level. The other covariates (Age, While working, and Musician) are not significant at the 0.05 level. The AIC (Akaike Information Criterion) for this model is 421.65. A lower AIC value indicates a better-fitting model.

#### 3.4.6 Confusion matrix

A confusion matrix is a popular performance evaluation tool used in machine learning and statistics to assess the performance of a classification model. It provides a summary of the model's predictions compared to the actual ground truth labels for a given dataset. The

confusion matrix is especially useful when dealing with problems where the output can belong to multiple classes or categories.

The confusion matrix is structured as a square matrix, with rows and columns representing the true classes and predicted classes, respectively. It consists of four main components:

**True Positives (TP):** The number of samples that were correctly predicted as positive (correctly classified instances of the positive class).

**True Negatives (TN):** The number of samples that were correctly predicted as negative (correctly classified instances of the negative class).

**False Positives (FP):** The number of samples that were incorrectly predicted as positive (instances of the negative class wrongly classified as positive).

**False Negatives (FN):** The number of samples that were incorrectly predicted as negative (instances of the positive class wrongly classified as negative).

The confusion matrix allows us to understand the performance of a classification model in terms of these four components, which helps in calculating various evaluation metrics like accuracy, precision, recall (sensitivity), specificity, and F1-score.

### 3.4.7 Interpretation

#### 3.4.7.1 Confusion matrix for first model

| Actual labels | Predicted labels |   |
|---------------|------------------|---|
|               | 0                | 1 |

|   |     |   |
|---|-----|---|
| 0 | 243 | 1 |
| 1 | 63  | 7 |

**True Positives (TP):** The number of samples correctly predicted as positive (actual 1, predicted 1) is 7.

**True Negatives (TN):** The number of samples correctly predicted as negative (actual 0, predicted 0) is 243.

**False Positives (FP):** The number of samples incorrectly predicted as positive (actual 0, predicted 1) is 1.

**False Negatives (FN):** The number of samples incorrectly predicted as negative (actual 1, predicted 0) is 63.

**Accuracy=** 0.793650793650794

**Precision=** 0.875

**Recall(Sensitivity)=**0.0985915492957746

**Specificity=**0.995901639344262

**F1-score:** 0.177215189873418

**Accuracy:** Accuracy is the proportion of correctly classified instances (both true positives and true negatives) out of all the instances in the dataset. An accuracy of 0.793650793650794 means that the model correctly predicts the outcome for approximately 79.37% of the data.

**Precision:** Precision is the proportion of true positive predictions (correctly predicted positive instances) out of all positive predictions (both true positives and false positives). A precision of 0.875 means that when the model predicts a positive outcome, it is correct approximately 87.5% of the time.

**Recall (Sensitivity):** Recall, also known as sensitivity, is the proportion of true positive predictions out of all actual positive instances in the

dataset. A recall of 0.0985915492957746 means that the model can identify approximately 9.86% of the actual positive instances correctly.

**Specificity:** Specificity is the proportion of true negative predictions (correctly predicted negative instances) out of all actual negative instances in the dataset. A specificity of 0.995901639344262 indicates that the model can identify approximately 99.59% of the actual negative instances correctly.

**F1-score:** The F1-score is the harmonic mean of precision and recall. It combines both metrics and provides a single score that balances precision and recall. An F1-score of 0.177215189873418 suggests that the model is not performing well in terms of both precision and recall.

The model has a high overall accuracy of 79.37%, meaning that it correctly predicts the outcome for approximately 79.37% of the data points.

The precision of 87.5% suggests that when the model predicts a positive outcome, it is correct around 87.5% of the time. This indicates that the model has a relatively low rate of false positives.

The recall (sensitivity) of 9.86% indicates that the model can identify only about 9.86% of the actual positive instances correctly. This means it has a high rate of false negatives, as it misses a large proportion of the positive cases.

The high specificity of 99.59% implies that the model has a very low rate of false positives, correctly identifying the negative instances.

The low F1-score of 0.177215189873418 indicates a poor balance between precision and recall. This suggests that the model is not performing well in capturing the positive instances while maintaining a low false positive rate.

### 3.4.7.2 Confusion matrix for second model

| Actual labels | Predicted labels |     |
|---------------|------------------|-----|
|               | 0                | 1   |
| 0             | 31               | 89  |
| 1             | 25               | 160 |

**True Positives (TP):** The number of samples correctly predicted as positive (actual 1, predicted 1) is 160.

**True Negatives (TN):** The number of samples correctly predicted as negative (actual 0, predicted 0) is 31.

**False Positives (FP):** The number of samples incorrectly predicted as positive (actual 0, predicted 1) is 89.

**False Negatives (FN):** The number of samples incorrectly predicted as negative (actual 1, predicted 0) is 25.

**accuracy=** 0.6262295

**Precision=** 0.642570281124498

**Recall(Sensitivity)=**0.864864864864865

**Specificity=**0.2583333333333333

**F1-score:** 0.737327188940092

**Accuracy:** Accuracy is the proportion of correctly classified instances (both true positives and true negatives) out of all the instances in the dataset. An accuracy of 0.6262295 means that the model correctly predicts the outcome for approximately 62.62% of the data.

**Precision:** Precision is the proportion of true positive predictions (correctly predicted positive instances) out of all positive predictions (both true positives and false positives). A precision of

0.642570281124498 means that when the model predicts a positive outcome, it is correct approximately 64.26% of the time.

**Recall (Sensitivity):** Recall, also known as sensitivity, is the proportion of true positive predictions out of all actual positive instances in the dataset. A recall of 0.864864864864865 means that the model can identify approximately 86.49% of the actual positive instances correctly.

**Specificity:** Specificity is the proportion of true negative predictions (correctly predicted negative instances) out of all actual negative instances in the dataset. A specificity of 0.258333333333333 indicates that the model can identify only about 25.83% of the actual negative instances correctly.

**F1-score:** The F1-score is the harmonic mean of precision and recall. It combines both metrics and provides a single score that balances precision and recall. An F1-score of 0.737327188940092 suggests that the model is reasonably good at finding the right balance between precision and recall.

The model has a moderate overall accuracy of 62.62%, meaning that it correctly predicts the outcome for approximately 62.62% of the data points.

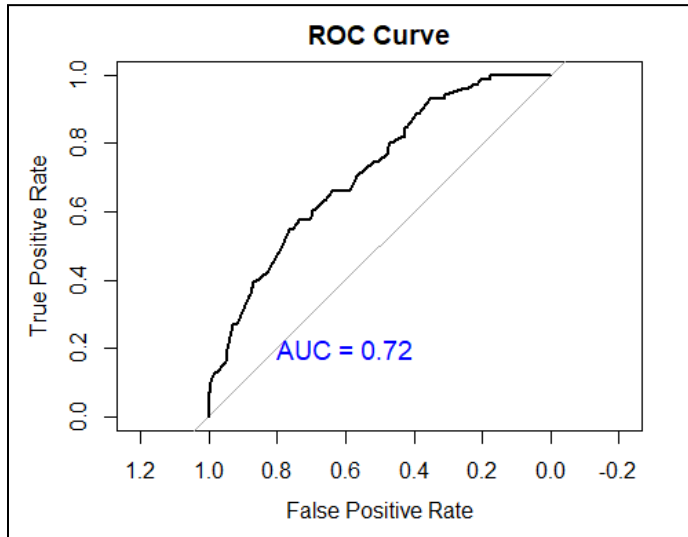
The precision of 64.26% suggests that when the model predicts a positive outcome, it is correct around 64.26% of the time. This indicates that there may be some false positives.

The recall (sensitivity) of 86.49% indicates that the model can identify approximately 86.49% of the actual positive instances correctly. This means it is relatively good at capturing positive cases.

The low specificity of 25.83% implies that the model has difficulty identifying actual negative instances, leading to a higher number of false positives.

The F1-score of 0.737327188940092 combines precision and recall and provides an overall measure of the model's performance. An F1-score close to 1 is desirable, and in this case, the model's F1-score suggests a reasonable performance in balancing precision and recall.

### 3.5 ROC and AUC curve



AUC - ROC curve is a performance measurement for the classification problems at various threshold settings. ROC is a probability curve and AUC represents the degree or measure of separability. It tells how much the model is capable of distinguishing

between classes. Higher the AUC, the better the model is at predicting 0 classes as 0 and 1 classes as 1. By analogy, the Higher the AUC, the better the model is at distinguishing between patients with the disease and no disease. The ROC curve is plotted with TPR against the FPR where TPR is on the y-axis and FPR is on the x-axis.

#### 3.5.1 ROC Curve for model 1

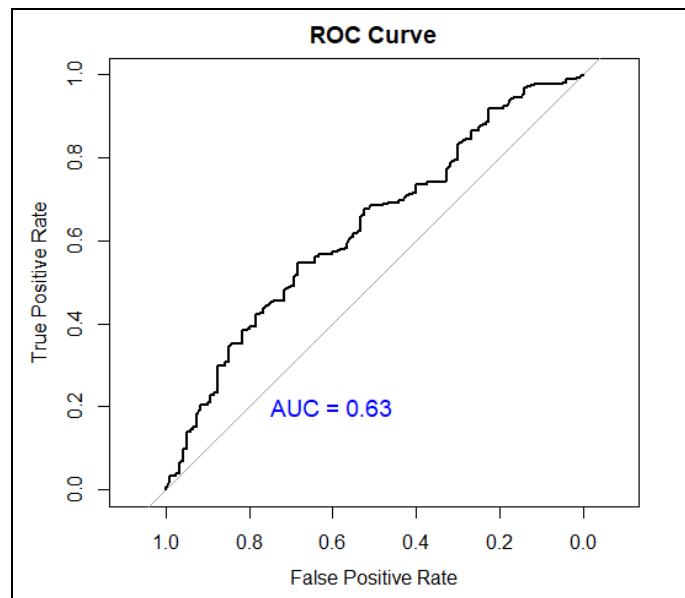
**Figure 24: ROC CURVE for model 1**  
**AUC =0.72**

### 3.5.2 ROC Curve for model 2

**Figure 25: ROC CURVE for model 2**  
**Here AUC=0.63**

#### **4.CONCLUDING** **REMARKED**

In this project, we attempted to draw a connection between the music preferences and the psychological well being of persons. As time passes people's psychological well being at risk ,they intend to search comfort and courage



to face reality which they can get from music as we know listening music or being indulge in music which gives comfort and a way to express our emotion.

While analyzing the dataset, Exploratory Data Analysis has seen that we got data on various age groups like Adults(above 25 years) ,Early adults(20-25 years) and Teenagers(15-19 years).Also female being musician is significant than the male being musician . Also there is a very significant number of females who don't even like to listen to Rock music. Also we can see that the number of people using music to remember previous memories is quite high. Also our dataset gives an interesting fact that males being emotional is higher than females also they are mostly introverted so they mostly listen to music to express their emotions.

We also did a chi square test of association to check if our observed data are associated or not .We can see that being a musician and having effect of music is not associated.Also we can see that being a classical music listener and having effect of music is associated. Also music effects and age groups are associated . we have found out that the opinions are dependent on only one criterion i.e, "listens classical song as their preference" and all other covariates i.e Age, Passion for music, listening classical while working do not affect their opinion.

We also analyze our dataset using ANOVA. We get that there is a significant difference between the age group (10-30) years and the age group (31-51) years.Also incase of age group (10-30) and the age group for above 52 years has significant difference so we can conclude that the younger generation are differ from the other age group due to their opinion.

Also we did 2 multiple logistic regression models. For the first logistic model the covariates are Age,Mother tongue,Zodiac sign,Language preference and we are using these covariates to predict the probability of being a musician.The accuracy of the first model is 79.37% and the

precision is 87.5%. And for the second model the covariates are Age ,listening to music while working,during working hours listen to devotional music and passionate for music or not and we are using this to predict the probability of music effect due to devotional music. This project gives an overview of how we use music as our secret weapon to survive this world.We can also make some project based on this topic where based on our mental and physical wellbeing we can predict what kind of music helps us to heal.

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